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Experimental and modeling study on the molecular weight distribution and properties of carbon particles in premixed sooting flames

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Abstract

The evolution of the molecular weight (MW) distribution and structural properties of carbon particulate formed in methane, ethylene and benzene fuel-rich premixed flames, burning in similar conditions of maximum flame temperature, was experimentally measured and modeled.

Both solubility and chromatographic separation of the carbon particulate allowed to follow the variation of molecular weight distributions of the different fractions. Structural properties of the carbon particulate in terms of H/C ratio and UV–visible absorption have been determined from particles inception to their maturation as a function of fuel identity. The experimental data were numerically predicted using a detailed gas-phase chemical kinetic mechanism coupled with a discrete sectional approach for the modeling of the molecule-to-particle process. The model was able to follow both particle concentration and structural properties for different fuels.

In the benzene flame high molecular mass aromatic compounds appeared very early in the oxidation region and were no longer detected at the end of flame whereas aliphatic fuels showed both aromatic molecules and freshly-nucleated soot particles persisting downstream of the flame. Elemental analysis and UV–visible properties of the carbon particulate showed that also the graphitization process is earlier and much faster in the benzene flame whereas in aliphatic flames the final ordered structure is produced later on the flame and with a slower rate. Detailed modeling of the carbon particulate yields and properties confirms the differences in the evolution of carbon particulate and its structure in benzene and aliphatic flames, also individuating the classes of species: molecules and clusters of molecules, responsible for soot inception.
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1. Introduction

The study of the fuel effect on combustion-formed carbon particulate from the inception up to final soot emission may provide clues for understanding the mechanism of soot formation.

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It has been assessed that fuel properties, affecting the surrounding hydrocarbons produced in the oxidation and pyrolysis phases of the combustion process, determine different soot yields [1–3]. Significant differences in the optical properties of condensed phases measured directly inside premixed flames of different hydrocarbons have been also found [2,4].

Advanced size- and structure-sensitive diagnostics, set up for characterizing carbon materials, have allowed to give deeper insights on the characteristics of particulate emission from practical combustion systems. Particle with sizes down to the nanometer range have been detected [5–8] and the structures of carbon particulates from the macroscopic length scale to molecular scale have been measured [9–11]. Modeling efforts for following in detail the particle nucleation and mass growth, have accompanied the progresses in soot diagnostics to include the variations of size and structural properties [12–17].

High-resolution transmission electron microscopy (HR-TEM), UV–visible and FT-IR spectroscopy results have recently shown significant differences in the chemical and structural characteristics of carbon particulate generated in flames of diverse fuels [18,19]. Molecular weight (MW)/size distribution of benzene soot has shown to shift toward large aggregates very early in the flame in respect to ethylene soot [20]. On the opposite, any changes in the soot volume fraction and in the bimodal feature of the size distribution have been found doping ethylene flames with benzene [21,22].

On-line sizing analysis as well as batch-sampling and diagnostic approaches applied to the total particulate are not able to infer the variations of structural properties of size-segregated fractions present in the wide MW/size distribution of the particulate from inception to maturation. This limits the comprehension of the chemistry and structural factors involved in soot formation as soot incepts and ages.

Size exclusion chromatography (SEC) of carbon particulate is an useful tool to analyze the size multimodality of soot, particularly in the nanometric size range (<20 nm) [20,23]. By using a UV–visible spectrometer as on-line chromatographic detector the SEC technique also allows to get spectroscopic parameters, sensitive to the size of the largest aromatic units constituting each size-segregated fraction [9,24].

In this work the separation/analysis capability of the SEC technique coupled with UV–visible detection has been applied to the total carbon particulate matter sampled in premixed flames in order to gain insights on the structural and MW variations occurring in the soot formation region of methane, ethylene and benzene flames.

The particulate concentration, MW distribution and composition have been numerically predicted by means of a detailed gas-phase chemical kinetic model coupled with a discrete sectional approach [16,17] for the modeling of the molecule-to-particle process.

2. Experimental

Premixed flames of methane, ethylene and benzene at atmospheric pressure were produced on a commercial McKenna sintered bronze burner (60 mm OD). The burner was water-cooled and the temperature of the cooling water was kept constant at 60 °C. An external shield of nitrogen was used to avoid air entrainment. Different cold-gas flow velocities, equivalence ratios and nitrogen dilution were used in order to obtain comparable flame temperatures as shown in Table 1, where the flame operating conditions are reported.

Flame temperatures were measured along the flame axis with a fast-response thermocouple (silica-coated 25 μm Pt/Pt–13%Rh) by using a fast-insertion procedure. A radiation correction procedure was applied to obtain corrected temperature profiles.

Combustion products, including gaseous, tarry and solid material, were collected from the flames by a water-cooled probe. Light C1–C6 hydrocarbons were on-line analyzed by an HP5890A gas chromatograph equipped with 7515 Chrompack Al2O3/KCl capillary column and flame ionization detector. Stable gases were analyzed on a HP5700A gas chromatograph with a 8700 Altech coaxial column and thermoconductivity detector.

Total particulate matter was recovered from the sampling line, constituted of an ice-trap and a Teflon filter, and extracted by dichloromethane (DCM) to separate the DCM-soluble species, containing polycyclic aromatic hydrocarbons (PAH) [9,25,26], from the DCM-insoluble species.

SEC for the MW evaluation of the particulate was carried out on a HP1050 High Pressure Liquid Chromatograph equipped with a Diode Array Detector for measuring the absorbance signal at a fixed wavelength and for measuring the on-line UV–visible spectra up to 800 nm. The on-line UV–visible spectra have been used for evaluating the optical band gap of each chromatographic peak. The absorbance signal of the chromatographic profiles was detected at $\lambda = 350$ nm that is the minimum wavelength where the interference of *N*-methylpyrrolidinone (NMP) solvent, used for the chromatographic elution, is avoided. Two different SEC columns, able to furnish a reliable MW distribution in a wide range of MW (100-1E10 u), were used. PL-gel styrene–divinylbenzene "individual pore" column (injection volume of 250 μl, flow rate 0.5 ml/min, 80 °C) was used to investigate the 10-1E5 u MW range [20] and a Jordi Gel divinylbenzene Solid Bead

Table 1
Flame characteristics.

Fuel	Cold gas velocity (v/cms)	C/O	ϕ	Fuel (vol.%)	O_2 (vol.%)	N_2 (vol.%)	T_{max} (K)
Methane	5.0	0.6	2.4	54.5	45.5	–	1700
Ethylene	4.0	0.8	2.4	44.4	55.6	–	1730
Benzene	4.0	0.77	1.9	4.9	19.0	76.1	1780

"non-porous" column (injection volume of 10 μl, flow rate 1 ml/min, room temperature) was used in the 1E5-1E10 u range [20,23]. The relation between the retention time and MW of polystyrene standards and carbonaceous materials of known dimensions, measured by dynamic light scattering, was used for the MW evaluation.

The MW distribution of the DCM-soluble fraction, analyzed on the "individual pore" column, generally shows the predominance of a sharp peak in the 100–400 u range and a much smaller peak in the range from 400 to 2000 u. A minor peak is also detected at MW >1E5 u [20].

The particulate, devoid of DCM-soluble species, was suspended in NMP by means of ultrasonic bath and filtered on 20 nm-porosity filter unit in order to separate a NMP-soluble fraction <20 nm from a NMP-insoluble fraction >20 nm.

The NMP-soluble fraction <20 nm (analyzed by the "individual pore" column) exhibited two peaks at 200–600 u (peaked at 400 u) and 600–5000 u (peaked at 800 u) [24]. A MW peak at MW >1E5 u was also detected [24].

The NMP-insoluble fraction >20 nm (analyzed by the "non-porous" column) showed a MW distribution peaked at about 1E10 u [20,23].

For the successive modeling work and in order to have a complete description of the multimode distribution of total particulate, the MW distribution has been re-composed taking into account for the contributions of the "solvent-soluble fraction" (DCM-soluble fraction + NMP-soluble fraction) and of the NMP-insoluble fraction >20 nm, here named "soot", to the total particulate.

It is worth to underline that the MW distribution obtained by SEC has just a comparative value since it has been obtained assuming the same absorptivity measured at 350 nm of the particulate fractions in the whole MW range.

H/C atomic ratio of the solvent-soluble fraction and dry soot were measured by a Perkin–Elmer 2400 CHNSO elemental analyzer.

3. Model summary

The model includes PAH formation and growth mechanisms and reactions responsible for molecular particle nucleation. The measured temperatures have been used as input for the model. Particle phase is based on a discrete sectional approach previously developed to take into account for both the carbon and hydrogen atom variations for each lumped species [16]. The model has been further implemented taking into account for the state of aggregation of the molecules. Indeed, in the version used in this work, three different types of species are considered: single molecules, clusters of molecules and aggregates of clusters. Details on the model are reported in [17].

Briefly, the model considers that molecular growth starts with the formation of radicals on aromatic molecules through unimolecular decomposition or bimolecular reactions with H or OH radicals. The addition of gas-phase compounds such as acetylene to aromatic radicals forms pericondensed PAH. Conversely, the addition of small aromatics forms incompletely-condensed aromatic compounds. Both pericondensed and incompletely-condensed aromatics can grow indefinitely forming extremely large mass molecules. It is also considered that recombination of radicals inhibits the growth process because of the reduction of radical concentrations. Reaction rates for the growth process are evaluated in analogy with gas-phase PAH reactions for the determination of the activation energy and by the gas kinetic theory for the evaluation of the collision frequencies. For reactions involving hydrogen atom abstraction, the effective hydrogen availability in the compound is also considered.

While pericondensed aromatics can only add acetylene or other gaseous hydrocarbons (if an aromatic molecule is added an incompletely-condensed aromatic is formed), an incompletely-condensed aromatic can undergo also dehydrogenation reactions forming pericondensed aromatic molecules. The dehydrogenation process does not affect the final concentration of the aromatic molecules, but it is important to take into account for this process and for correctly considering the effect of the chemical structure of aromatics on further molecule/particle evolution.

The H/C ratio of the aromatic molecules provides an approximate estimation of the pericondensed and incompletely-condensed structure of aromatics. For a given C atom number, lower H/C ratio characterizes the class of pericondensed aromatics whereas high H/C ratios characterizes incompletely-condensed aromatics. Both pericondensed and incompletely-condensed aromatics constitute the so-called Type 1 class.

Parallel to the molecular growth process, high mass aromatic molecules can form clusters

of two (dimers) aromatic units through dispersion forces. Coalescence with other molecules/clusters causes further clustering. All clusters constitute the Type 2 class. The negative effect of the temperature on clustering process due to rebound of the coalescing species has been considered. Clustering is enhanced by the increase of the molecular mass and of the aromaticity of the species. Both effects are taken into account. The increase of the molecular mass is considered through the evaluation of the change of the collision rate with species size; the increase of species aromaticity is accounted for by the Hamacker constant.

As the size of Type 2 class increases, colliding molecules/clusters are no longer incorporated, but aggregate. The characteristic coalescence time for Type 2 class has been calculated following the procedure reported in [27]. Clusters maintain their shapes and further aggregate forming the Type 3 class. Type 3 compounds can grow due to coalescence of molecules from the surrounding gas-phase (Type 1) or collisions with other clusters (Type 2).

It has been considered that, concurrently with molecular growth, clustering and aggregation processes, oxidation by OH and O_2 also occurs.

4. Results and discussion

Methane, ethylene and benzene fuel-rich premixed flames have been investigated to study the effect of the fuel structure on particulate formation and on the chemical and morphologic characteristics of the particles. Different flame conditions, reported in Table 1, have been used to keep the flame temperature in a narrow range, between 1700 and 1780 K.

To take into account for the different fed carbon and flame dilutions, experimental data have been expressed as carbon yields and compared with model predictions considering the separation of the total particulate in the solvent-soluble fraction, constituted of compounds with MW $<10^5$ u and soot fraction, i.e. particles with MW $>10^5$ u.

Figure 1a reports the carbon yields of the total particulate, solvent-soluble fraction and soot fraction measured along the axis of methane, ethylene and benzene flames. The particulate yield increases along the axis of all flames, but a higher formation rate can be observed in the benzene flame. However, in spite of the higher propensity of benzene to form soot, the particulate yield is lower in respect to the ethylene flame due to the higher dilution of the benzene flame (Table 1).

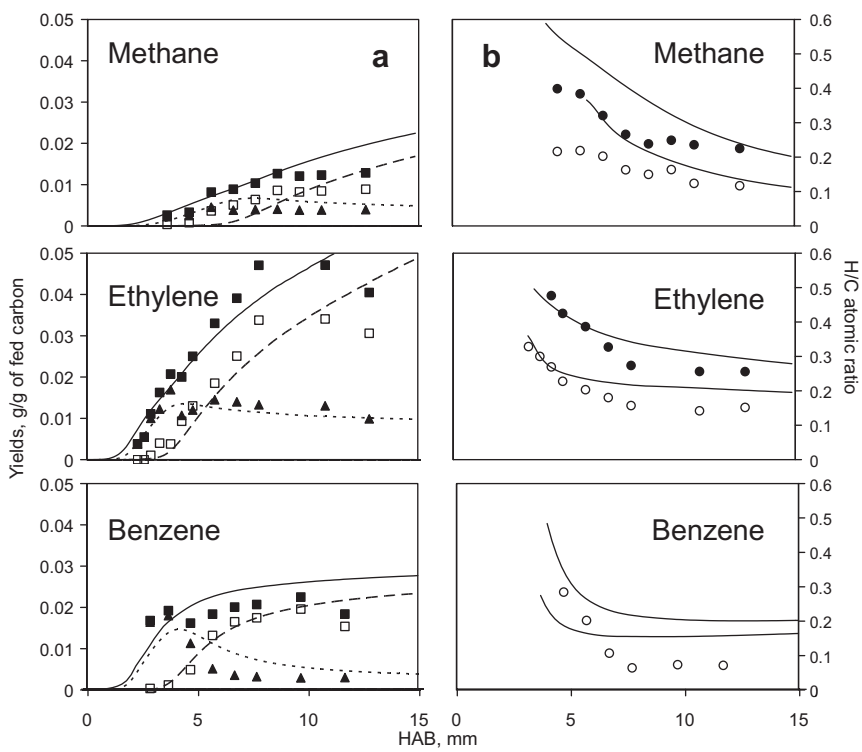


Fig. 1. (a) Yields of total particulate matter (■ experiments, __ modeling), soot (□ experiments, __ __ modeling) and solvent-soluble species (▲ experiments, - - - modeling) for the three flames. (b) H/C ratio of total particulate (● experiments, heavy line for modeling) and soot (○ experiments, thin line for modeling) for the three flames.

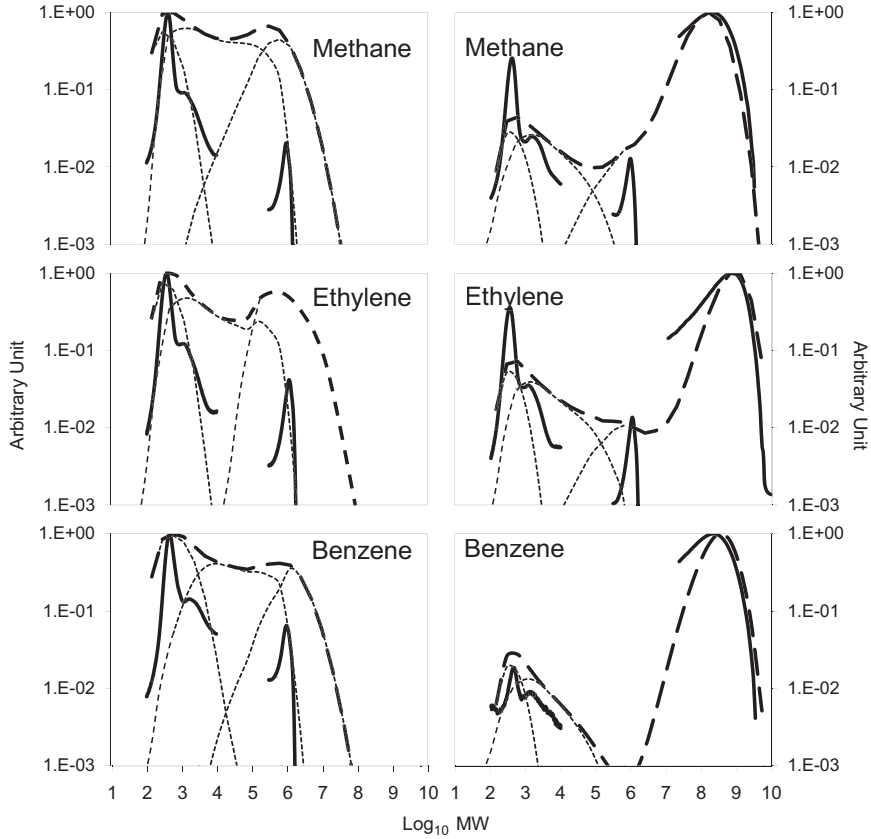


Fig. 2. Normalized SEC chromatograms (__) compared with normalized concentration profile of total particulate (__ __) and type1, type2 and type3 compounds (- - -) (see text) at inception (left part) and in the post-flame region (right part) of the three flames.

The total particulate rise is initially due to the increase of the solvent-soluble fraction yield. After the maximum, the steep decrease of the solvent-soluble fraction occurs in the benzene flame, reaching very low values as soot production rapidly levels-off. Along the ethylene and methane flames the solvent-soluble fraction increases and, after the maximum, it remains quite constant whereas soot exhibits a monotonic increase throughout the flames, although with a lower rate in the final flame region.

Model predictions, also reported in Fig. 1a, fairly well reproduce the profile shapes and values of yields, especially for ethylene and benzene flames.

The experimental and modeled H/C ratios profiles of total particulate and soot fraction are reported in Fig. 1b to follow the chemical and thermal histories of the carbon particulate. The H/C ratio of the total particulate in the benzene flame could not be evaluated due to the interference of oxygenated aromatic species in the solvent-soluble fraction.

The H/C ratio decrease of particulate along the flame axis testifies the occurrence of dehydrogenation/graphitization process leading to the increase of the aromatic carbon network. The model fairly reproduces this trend, although the H/C ratios are overestimated for the methane and benzene flames. It is worth to remind that the H/C ratio decrease is mainly due to soot fraction rather than to solvent-soluble fraction [26], demonstrating that the increase of the extension of the aromatic network primarily occurs in soot particles. H/C ratio of benzene soot rapidly and early decreases reaching very low values downstream of the flame, where the dehydrogenation process ends. The model is able to reproduce well the fast decrease of soot H/C ratio in the benzene flame.

The different behaviors of the H/C ratio along the three flames can be ascribed to the different flame environments in which particles nucleate and graphitize. Indeed, particulate formation occurs already in the main oxidation region of the benzene flame, being the flame front, i.e. the maximum concentration of H and OH atoms,

located at about 3.5 mm (Fig. 1a). The high temperature and the high radical concentration favor molecular growth of aromatics and particle inception. At the same time they promote dehydrogenation and condensation reactions leading to the formation of mature soot particles already upstream of the flame front. Conversely, particulate formation in the aliphatic fuel flames occurs downstream the flame front, which is located at about 2 mm in both methane and ethylene flames. Here the temperature is slightly decreasing while the concentrations of H and OH radicals attain the equilibrium values. The temperature and radical concentration decrease inhibits the dehydrogenation causing a reduced and delayed maturation of soot particles which keep on a higher hydrogen fraction.

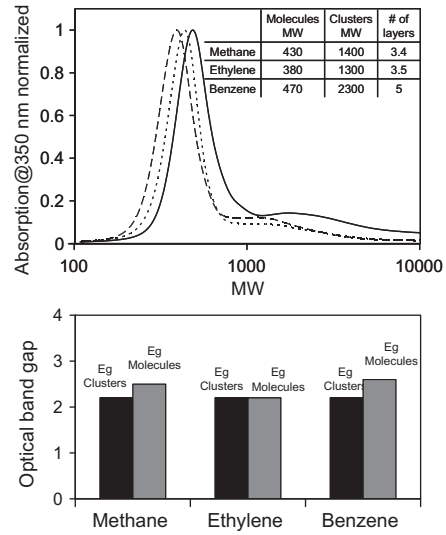
Further insights on the particle structure evolution have been obtained by evaluating the MW distribution of particulate matter. Figure 2 shows the normalized mass distributions of total particulate experimentally determined at the inception (left side of Fig. 2) and in the post-flame region (right side of Fig. 2) of the three flames in comparison with the model results. It is worth to remind that the MW distribution of the total particulate has been obtained by re-composition of the SEC chromatograms of the solvent-soluble fraction and soot assuming the same absorptivity measured at 350 nm of the particulate fractions.

Overall, the MW distributions show four peaks belonging to species in the 100–1000 u range, in the 1000–5000 u range, in the 10^5–10^6 u range and in the 10^6–10^{10} u range. At particle inception the first three peaks, mostly belonging to the solvent-soluble fraction, are predominant whereas in the post-flame region the higher MW peak in the 10^6–10^{10} u range, due to soot, prevails.

Model results are in good agreement with experimental data and are useful in giving information on the nature of the very high MW classes, not amenable to chemical analysis. The modeled first peak derives from aromatic molecules having masses up to 1000 u (Type 1) in agreement with the mass spectrometric analysis [24]. On the basis of the model prediction, the second peak can be ascribed to clusters of aromatic molecules with MW up to about 5000 u (Type 2), whereas the third peak is due to clusters of about 10^6 u (Type 2). These clusters constitute the first species which have acquired a solid nature and cannot longer coalesce with other molecules, but can only agglomerate. The fourth peak is due to agglomerates of these clusters (Type 3) constituting soot.

In light of this general picture inferred through the modeling, the main differences occurring in the MW region up to 10^6 u in terms of peak position and relative abundance can be noticed.

In particular it can be noted that the modes of MW distribution in the MW region up to 10^6 u related to Type 1 molecules and Type 2 clusters



	Molecules MW	Clusters MW	# of layers
Methane	430	1400	3.4
Ethylene	380	1300	3.5
Benzene	470	2300	5

Fig. 3. Top: Normalized SEC chromatograms up to 10,000 u at inception for methane (- - -), ethylene (_____) and benzene (__) flames. Bottom: Optical band gap (Eg) of molecules and clusters separated on the base of SEC analysis at inception in the three flames.

are still visible at the end of ethylene and methane flames and practically negligible in the benzene flame (left part of Fig. 2). Model well reproduces this behavior showing a persistent significant presence of modes around 300, 1500 and 10^6 u at the end of ethylene and methane flames.

To analyze the passage from Type 1 molecules to Type 2 clusters occurring at soot inception, the optical band gap analysis, sensitive to the mean molecular mass of the largest aromatic units constituting each class [28–32], has been carried out. The optical band gap has been evaluated on the on-line UV–visible spectra measured on the apex of Types 1 and 2 peaks detected in the MW distribution up to 10^4 u of particulate at inception.

Figure 3 reports for all flames the MW distribution up to 10^4 u of particulate at inception (upper part of Fig. 3) together with the optical band gaps (Eg) of the Type 1 molecule peak and of the Type 2 cluster peak (lower part of Fig. 3).

It is noteworthy that in the three flames the Eg values of molecules and clusters are quite similar at soot inception. This demonstrates that the aromatic units in molecules are the aromatic units constituting the clusters. It turns out that the ratio of the mean MW of clusters and molecules is representative of the number of molecules staked in the cluster, i.e. the number of stacked layers. The inset in Fig. 3 reports the MW of molecules and clusters, evaluated on the peak maximum, and the calculated number of stacked layers. A low degree of stacking can be noted in the methane and ethylene flames whereas the number of

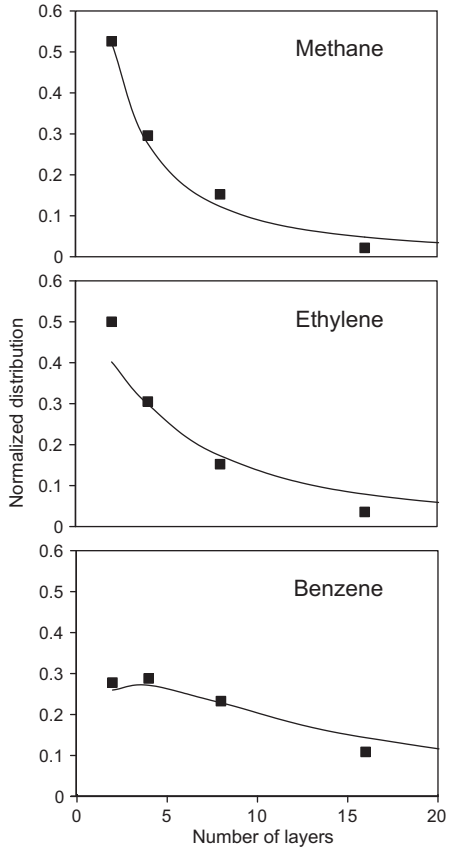


Fig. 4. Experimental (dots) and modeled (lines) normalized distribution of number of stacked layers per cluster at inception in the three flames.

stacked layers in the cluster is higher in the benzene flame.

Considering the wideness of MW peaks, a detailed evaluation of the number distribution of stacked layers at particle inception in the three flames can be obtained by deconvolution of MW distribution, reported in the upper part of Fig. 3, into two separate distribution functions of molecules and clusters. The ratio of the distribution functions of clusters and molecules turns into the final distribution of the number of stacked layers per cluster reported in Fig. 4. Modeling results in terms of the ratio of distribution functions of Types 1 and 2 classes are also reported in Fig. 4. Model and experimental data are in very good agreement both indicating that the stacked layers number is much lower in methane and ethylene flames in respect to the benzene flame. This means that a relevant higher level of stacking/organization of the precursor particles is obtained at inception in the benzene flame. The more ordered structure of early soot in benzene flame, is consistent with its lower H/C ratio, and causes the higher soot formation rate and the peculiar evolution of benzene particulate properties in terms of size and oxidation reactivity showed in previous work [18–20].

5. Conclusions

The MW distribution of carbon particulate has been experimentally measured and modeled in methane, ethylene and benzene premixed flames. In the benzene flame aromatic molecules and nascent soot particles appeared very early in the oxidation region and were no longer detected at the end of flame whereas aliphatic fuels showed both aromatic molecules and freshly-nucleated soot particles persisting downstream of the flame. Elemental analysis and UV–visible properties of the carbon particulate showed that also the graphitization process is earlier and much faster in the benzene flame whereas in aliphatic flames a less ordered structure is produced later on the flame and with a slower rate. Detailed modeling of the carbon particulate yields and properties confirms the differences in the evolution of carbon particulate and its structure in benzene and aliphatic flames, also individuating the classes of species: aromatic molecules and clusters of aromatic molecules responsible for soot inception.

Nascent soot particles in benzene flame, earlier formed in the main oxidation zone overlapped with the pyrolysis region of the flame, exhibiting a more organized/stacked aromatic structure. Here graphitization is enhanced by the high radical concentrations typical of the flame front. As a consequence the coagulation is also enhanced for effect of the increased aromaticity (lower H/C ratio) of the incipient particles which result in increasing binding energies of the formed clusters.

The inception of particles in the aliphatic flames occurs in the pyrolysis region that is well distinct from the main oxidation region and located downstream of the flame front. Here the temperature is slightly decreasing while the concentrations of small radicals are attaining the equilibrium values. Both the temperature and radical concentration decreases, turn into the formation of less organized structures subjected to a slower and lower evolution toward larger mature soot particles in the methane and ethylene flames.

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